



HydroGraphRAG: Intelligent GIS Assistant

Geospatial code generation and visualization based on knowledge graphs.

Enter hydrology query:

What is flood risk in Almaty City?

Generate solution

✓ Solution generated!

Category

IMPLICIT

Triples extracted

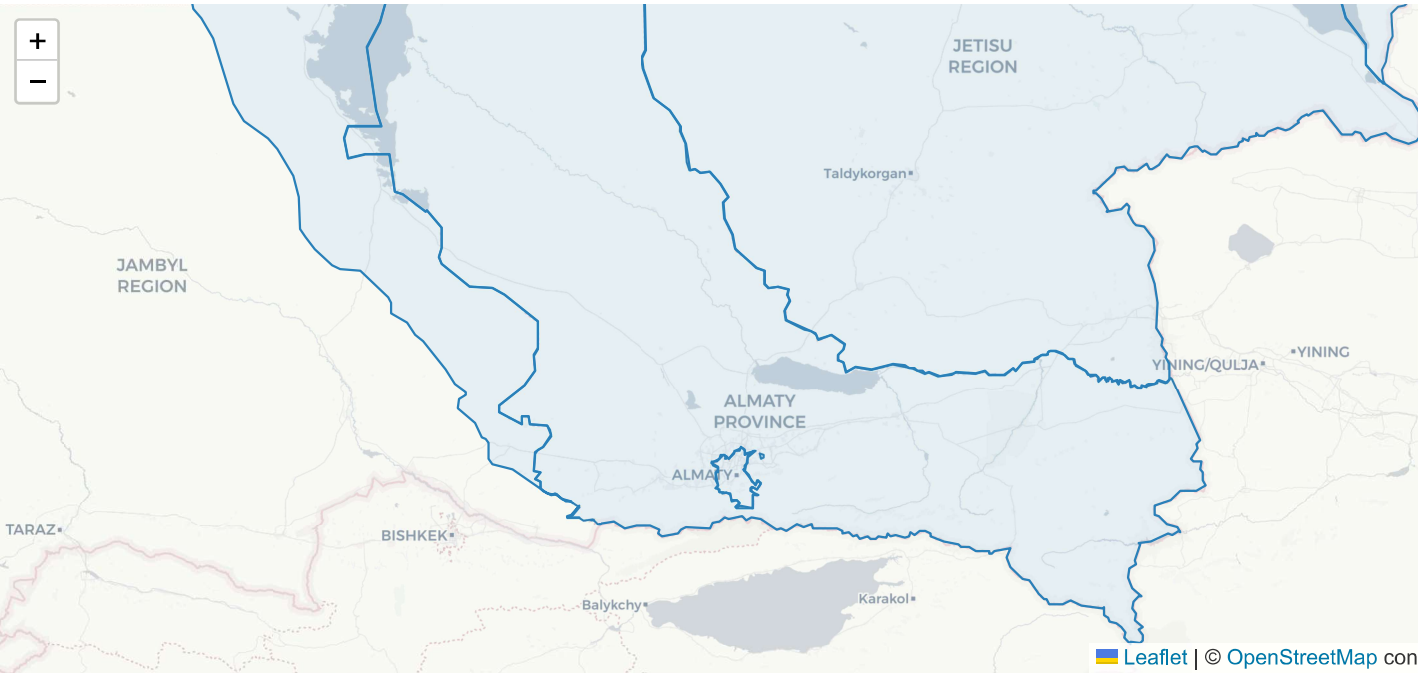
379

Points on map

0



Geospatial Visualization



Analytical Report and Code

Modeling Solution:

- **Almaty City:** Located in the region of Almaty City.
- **Kishi Almaty River:** A feature of interest within Almaty City.
- **Water Quality Class:** Class 3, indicating moderate water quality.
- **Arable Land Area:** 19419.5 hectares.
- **Date and Time:** Data recorded on 2020-01-01T00:00:00.

Implementation Code:

```
import geopandas as gpd
import folium
from shapely import wkt

# 1. Load basin
basin_data = gpd.read_file(r"data/basin_data.shp").to_crs('EPSG:4326')
centroid = basin_data.geometry.centroid.iloc[0]

# 2. Init map
m = folium.Map(location=[centroid.y, centroid.x], tiles='CartoDB positron', zoom_start=8)
folium.GeoJson(basin_data.to_json(), style_function=lambda x: {'fillColor': 'green', 'color': 'darkgreen', 'fillOpacity': 0.5})

# 3. Add points here based on Graph Knowledge
points = [
    {"name": "Kishi Almaty River", "wkt": "POINT(76.0 43.0)"}
]

# 4. Draw markers
for p in points:
    geom = wkt.loads(p["wkt"])
    folium.Marker(location=[geom.y, geom.x], popup=p["name"]).add_to(m)

m.save("api_1780740528.html")
```